Aerosol Can Coating Calculations

Corrected BOM Consumption Report — High-End Estimates

Can Dimensions: 45mm outer diameter × 160mm height, wall thickness ~0.12mm

Step 1: Calculate the Can Surface Area

To determine the required coating quantity per can, the surface area must be calculated from the physical geometry of the cylinder. The calculations are split into external and internal surfaces due to the differing application methods and regions coated.

External Surface Area (for Base Coat & Varnish)

The outside of the can is coated on the side wall only. The bottom is left uncoated as it sits on the conveyor line during processing. The side wall of the cylinder is represented by the formula:

```
Area_outer = π × diameter × height = π × 45 mm × 160 mm = 22,619 mm² = 0.02262 m²
```

Internal Surface Area (for Internal Lacquer)

The interior of the can is coated on both the side wall and the bottom base to protect the aluminum against corrosion from the chemical formulation of the aerosol product. The wall thickness is 0.12 mm, meaning the inner diameter is 44.76 mm (45 mm minus 2 × 0.12 mm).

```
Area_inner_wall = π × 44.76 mm × 160 mm = 22,499 mm²
Area_inner_base = π × (22.38 mm)² = 1,573 mm²
Area_inner_total = 22,499 mm² + 1,573 mm² = 24,072 mm² = 0.02407 m²
```

Sanity Check

These surface area measurements are small. A single can has an external side wall surface area equal to approximately 1/44th of a square meter, and an internal surface area of approximately 1/42nd of a square meter. This is consistent with a standard 45mm × 160mm aluminum aerosol container.

Step 2: Multiply Surface Area by TDS Wet Film Weight

The Technical Data Sheets (TDS) specify the recommended wet film weight applied in grams per square meter (g/m²). To calculate the theoretical consumption per can, we multiply the surface area by the wet film weight. To ensure high-end coverage and prevent under-application, the high-end value of the TDS range is selected for all coating items.

```
Consumption per can = Surface area (m²) × Wet film weight (g/m²)
```

Internal Lacquer — SCHEKOSOL INT PROT Gold (400 9 901)

TDS Property / Specification	Value
Wet film weight range	37.0 – 44.0 g/m²
Solid content	36%
Application method	Internal Spraying

Wet Film Option	Calculation	Per Can	Per 1000 Cans (Theoretical)
Low (37.0 g/m²)	0.02407 m² × 37 g/m²	0.891 g	0.891 kg
Midpoint (40.5 g/m²)	0.02407 m² × 40.5 g/m²	0.975 g	0.975 kg
High End (44.0 g/m²) ★	**0.02407 m² × 44.0 g/m²**	**1.059 g**	**1.059 kg**

Base Coat — SCHEKOSOL WH BC (422 0 903)

TDS Property / Specification	Value
Wet film weight range	32.0 – 43.0 g/m²
Solid content	71%
Application method	Roller Coating

Wet Film Option	Calculation	Per Can	Per 1000 Cans (Theoretical)
Low (32.0 g/m²)	0.02262 m² × 32 g/m²	0.724 g	0.724 kg
Midpoint (37.5 g/m²)	0.02262 m² × 37.5 g/m²	0.848 g	0.848 kg
High End (43.0 g/m²) ★	**0.02262 m² × 43.0 g/m²**	**0.973 g**	**0.973 kg**

Varnish — SCHEKOSOL OPV GLOSSY (422 9 900)

TDS Property / Specification	Value
Wet film weight range	13.0 – 19.0 g/m²
Solid content	43%
Application method	Roller Coating

Wet Film Option	Calculation	Per Can	Per 1000 Cans (Theoretical)
Low (13.0 g/m²)	0.02262 m² × 13 g/m²	0.294 g	0.294 kg
Midpoint (16.0 g/m²)	0.02262 m² × 16 g/m²	0.362 g	0.362 kg
High End (19.0 g/m²) ★	**0.02262 m² × 19 g/m²**	**0.430 g**	**0.430 kg**

Step 3: Add Process Losses

The quantities calculated in Step 2 represent the theoretical wet film thickness applied directly to the container. In a real-world manufacturing environment, additional raw materials are consumed due to process losses (e.g., overspray in narrow cans, line startup/stop waste, roller cleaning, and color changeovers). The BOM must incorporate these losses to prevent inventory shortages.

Coating Application Method	Transfer Efficiency	Process Loss (Waste)	Notes / Industry Context
Roller Coating (External)	90% – 95%	5% – 10% loss	Highly efficient roller lines, waste mainly from cleanup and startup.
Internal Spray (Airless)	60% – 70%	30% – 40% loss	Standard for aerosol spray lines. Overspray occurs at container mouth.
Internal Spray (Electrostatic)	70% – 85%	15% – 30% loss	Modern lines with electrical charging to draw paint particles.

Technical Note on Solid Content

The wet film weights specified in the TDS (e.g., 32–43 g/m² for the base coat) are already formulated by the supplier to achieve the target dry film thickness (10–12 µm) given the product's solid content (71% for base coat).

Therefore, we **do not** divide the wet film weight by the solid percentage again. Doing so would lead to double-counting and artificially inflate consumption values.

Step 4: Recommended BOM Values (High-End Spec)

Applying conservative industry-standard waste factors (10% process loss/90% transfer efficiency for roller coating; 35% process loss/65% transfer efficiency for interior spray coating) to the high-end theoretical consumption numbers yields the following recommended BOM quantities:

Internal Lacquer (400 9 901)

Applying a 35% spray loss factor to the high-end wet film weight:

```
BOM Value = 1.059 g / 0.65 = 1.629 g/can = 1.63 kg/1000 cans
```

Base Coat (422 0 903)

Applying a 10% roller loss factor to the high-end wet film weight:

```
BOM Value = 0.973 g / 0.90 = 1.081 g/can = 1.08 kg/1000 cans
```

Varnish (422 9 900)

Applying a 10% roller loss factor to the high-end wet film weight:

```
BOM Value = 0.430 g / 0.90 = 0.478 g/can = 0.48 kg/1000 cans
```

Alternative Coating Products (High-End Estimates)

The calculations below apply the same methodology to the alternative coatings listed on the TDS and added to the BOM. These estimates likewise assume the highest end of the TDS wet film ranges and standard process losses.

SCHEKOSOL INT BEIGE (400 4 902) — Internal Lacquer Alternative

- **TDS Specification:** 41.0 – 48.0 g/m² wet film weight, 38% solids, internal spray application.
- **High-End Spec:** 48.0 g/m²
- **Theoretical Consumption:** 0.02407 m² × 48.0 g/m² = 1.155 g/can (1.155 kg/1000 cans)
- **BOM Value (with 35% spray loss):** 1.155 g / 0.65 = 1.777 g/can = **1.78 kg/1000 cans**

SCHEKOSOL CLEAR BC (422 9 918) — Base Coat Alternative

- **TDS Specification:** 25.0 – 31.0 g/m² wet film weight, 51% solids, roller application.
- **High-End Spec:** 31.0 g/m²
- **Theoretical Consumption:** 0.02262 m² × 31.0 g/m² = 0.701 g/can (0.701 kg/1000 cans)
- **BOM Value (with 10% roller loss):** 0.701 g / 0.90 = 0.779 g/can = **0.78 kg/1000 cans**

SCHEKOSOL OPV SILKMATT (422 9 903) — Varnish Alternative

- **TDS Specification:** 11.0 – 17.0 g/m² wet film weight, 47% solids, roller application.
- **High-End Spec:** 17.0 g/m²
- **Theoretical Consumption:** 0.02262 m² × 17.0 g/m² = 0.385 g/can (0.385 kg/1000 cans)
- **BOM Value (with 10% roller loss):** 0.385 g / 0.90 = 0.428 g/can = **0.43 kg/1000 cans**

Complete Corrected BOM Summary

Below is the consolidated bill of materials showing both the original estimates and the corrected, high-end coating values based on precise container dimensions and process loss adjustments.

#	Material Name	Material Group	Old Value (kg/1000)	Corrected Value (kg/1000, High End)
1	45mm Aluminum Slug	Slug	20.000	20.000 (No change)
2	LUBRIMET GR8	Lubricant	0.021	0.021 (No change)
3	ALULIQUID 13	Cleaner	5.000	5.000 (No TDS rate — keep as-is)
4	SCHEKOSOL INT PROT Gold (400 9 901)	Lacquer	4.600	**1.630**
5	SCHEKOSOL WH BC (422 0 903)	Base Coat	4.000	**1.080**
6	SunAltec OPAQUE WHITE (Ink)	Ink	0.300	0.300 (No change)
7	SCHEKOSOL OPV GLOSSY (422 9 900)	Varnish	2.000	**0.480**
8	SCHEKOSOL INT BEIGE (400 4 902)	Lacquer	5.100	**1.780**
9	SCHEKOSOL CLEAR BC (422 9 918)	Base Coat	3.000	**0.780**
10	SCHEKOSOL OPV SILKMATT (422 9 903)	Varnish	1.500	**0.430**

Assumptions & Caveats

- **Can height:** Assumed at exactly 160 mm. If the final container is shorter or taller, all surface area and coating consumption calculations must be updated proportionally.
- **Internal Spray Loss (35%):** Assumed for internal spray application in narrow monobloc aerosol cans. Real-world transfer efficiency varies depending on the specific nozzles, pressures, and nozzle insertion depth used on your production line.
- **Roller Loss (10%):** Standard waste estimate for external roller coater lines. Well-optimized, high-speed lines may achieve waste percentages as low as 5%.
- **Number of Coats:** Calculations are based on **1 single coat** of each material. Note that the TDS for Internal Lacquer mentions: 'Recommended procedure: 1 or 2x'. If product specifications dictate a double internal coat (2x application), the internal lacquer BOM quantity must be doubled (e.g., from 1.63 kg/1000 to 3.26 kg/1000).
- **Production Verification:** All corrected theoretical values should be audited and adjusted against actual line trials and chemical consumption weights once manufacturing begins.